Clustering Results

CA Assignment 2 · Clustering Algorithms

Algorithm: Agglomerative Hierarchical Clustering (AHC)

1. Clustering Algorithm: Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering (AHC) is a bottom-up clustering algorithm that builds a hierarchy of nested partitions by iteratively merging the two closest clusters. The process begins with N singleton clusters — one per data point — and terminates when all points belong to a single cluster. A flat partition into K clusters is obtained by cutting the resulting dendrogram at the level that produces K groups. The cut point is selected by maximising the silhouette coefficient over $K = 1, \ldots, 10$.

1.1 Algorithm Mechanics

At each iteration, AHC computes the inter-cluster distance for all current cluster pairs using a linkage criterion, then merges the pair with the smallest distance. The choice of linkage criterion fundamentally determines the geometry of the resulting clusters:

- **Single linkage:** $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. Prone to chaining; produces elongated, non-compact clusters.
- **Average linkage:** $d(A, B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$. Less prone to chaining than single linkage but still susceptible on text data where one domain greatly outnumbers others.
- **Ward linkage (used in this work):** merges the pair (A, B) that minimises the increase in total within-cluster sum of squares. Produces compact, balanced, well-separated clusters.

1.2 Why AHC?

- **No assumption about cluster shape.** Unlike K-means (MacQueen, 1967), which assumes spherical, equally-sized clusters, AHC can recover clusters of arbitrary shape and size. The dataset exhibits strong size imbalance: Cluster 2 (Shakespeare, 790 documents) and Cluster 3 (Wikipedia, 90 documents).
- **Deterministic output.** AHC produces the same result on every run, unlike K-means which depends on random centroid initialisation. This is important for reproducibility.
- **No need to pre-specify K.** AHC builds the complete dendrogram in a single pass; the optimal K is selected post-hoc by silhouette analysis across $K = 1$ to 10.
- **Literature support.** AHC is a standard and well-validated algorithm for text clustering with dense embeddings. Petukhova et al. (2024) include AHC as one of their primary benchmark algorithms and report competitive performance with BERT-family embeddings.

2. Configuration: Ward Linkage and Euclidean Distance

2.1 Ward Linkage

Ward linkage (Ward, 1963) selects the merge that minimises the increase in total within-cluster sum of squares (WCSS). Formally, the cost of merging clusters A and B is:

$$\Delta(A,\, B) \;=\; \frac{|A|\,|B|}{|A| + |B|}\, \|\boldsymbol{\mu}_A - \boldsymbol{\mu}_B\|^2$$

where $\boldsymbol{\mu}_A$ and $\boldsymbol{\mu}_B$ are the cluster centroids and $|A|$, $|B|$ are cluster sizes. Ward is the only linkage criterion that directly optimises cluster compactness at every merge step. This produces balanced, well-separated clusters and is robust against the **chaining problem** — the tendency of average and single linkage to greedily merge individual points into one large elongated cluster rather than forming distinct groups.

On this dataset, empirical investigation confirmed the chaining problem: average linkage consistently collapsed the clustering to $K = 2$ regardless of metric, absorbing the Wikipedia sub-corpus into the larger modern English cluster and ignoring its thematic distinctiveness. Ward linkage recovered the semantically correct $K = 3$ solution.

2.2 Euclidean Distance

Ward linkage is mathematically defined *only* for Euclidean distance — the formula for $\Delta(A, B)$ above uses squared Euclidean norms between centroids. Applying Ward with any other metric (e.g., cosine) is not mathematically valid and produces undefined behaviour.

The use of Euclidean distance is appropriate here for two reasons. First, L2 normalisation is applied to the raw embeddings before PCA. For unit vectors in $\mathbb{R}^{384}$, Euclidean and cosine distances are monotonically equivalent:

$$d_{\text{Euclidean}}(\hat{\mathbf{a}},\, \hat{\mathbf{b}})^2 = 2\big(1 - \cos(\hat{\mathbf{a}},\, \hat{\mathbf{b}})\big)$$

PCA of unit vectors preserves this angular geometry in the principal component subspace, so Euclidean distance in PCA space remains semantically meaningful. Second, the PCA output is deliberately left un-normalised, so Euclidean distances in the 100-dimensional PCA space weight the earlier, higher-variance principal components more heavily — since these components capture the most semantically significant axes of variation, this variance weighting is a useful inductive bias for cluster discovery.

3. Silhouette Analysis

The silhouette coefficient is an **internal** cluster validity index that quantifies how well each data point fits its assigned cluster relative to the nearest alternative cluster (Rousseeuw, 1987). For data point i:

$$a(i) = \text{mean intra-cluster Euclidean distance from } i \text{ to all other points in its cluster}$$

$$b(i) = \text{mean Euclidean distance from } i \text{ to all points in the nearest other cluster}$$

$$s(i) = \frac{b(i) - a(i)}{\max\big(a(i),\, b(i)\big)} \;\in\; [-1,\, +1]$$

The overall silhouette score is:

$$S = \frac{1}{N}\sum_{i=1}^{N} s(i)$$

A score near +1 indicates well-separated, compact clusters; near 0 indicates overlapping clusters; negative values indicate misassignment. Both $a(i)$ and $b(i)$ are computed using the same Euclidean metric used for clustering, ensuring a fully consistent and internally valid evaluation.

3.1 Results for $K = 1$ to 10

K	Silhouette Coefficient	Notes
1	N/A	Undefined — requires at least 2 clusters
2	0.084815	
3	**0.091923**	**Global optimum (peak)**
4	0.070843	
5	0.045152	
6	0.035833	
7	0.039736	
8	0.041355	
9	0.045331	
10	0.050710	

Table 1: Silhouette coefficients for AHC (Euclidean, Ward) over $K = 1$ to 10. $K = 3$ is highlighted as the global optimum.

4. Optimal Value of K and Justification

4.1 Quantitative Justification: Silhouette Analysis

The silhouette analysis unambiguously identifies $\mathbf{K = 3}$ as the optimal number of clusters, with a peak coefficient of **0.091923**. The score at $K = 3$ exceeds that at $K = 2$ (0.084815) and all higher values of K, which decrease to a minimum of 0.035833 at $K = 6$ before gradually recovering from $K = 7$ onwards. The shape of the curve — a clear peak at $K = 3$ followed by sustained decline — is a strong quantitative signal that the data contains three natural groupings at the granularity captured by the MiniLM embedding and the 100-dimensional PCA feature space.

4.2 Qualitative Justification: Semantic Cluster Interpretation

The three clusters produced at $K = 3$ have clear, interpretable semantic identities:

The merging of reviews and news into a single cluster (Cluster 1) is semantically well-founded: both sub-corpora are written in contemporary English, use similar vocabulary (named entities, everyday objects, locations), and share syntactic patterns characteristic of modern prose. In MiniLM's embedding space, their angular separation is smaller than their collective angular separation from Shakespeare. This is not a failure of the algorithm but an **accurate reflection of the linguistic properties of the data**.

Cluster	Size	Domain	Linguistic Characteristics
1	880 (50.0%)	Reviews + News	Modern English prose: restaurant and business reviews (Yelp-style) and CNN-style news articles. Informal to formal register, contemporary vocabulary, first and third person narration.
2	790 (44.9%)	Shakespeare (Coriolanus)	Early Modern English: play dialogue from Coriolanus. Archaic vocabulary, poetic metre, character prefixes (e.g., CORIOLANUS:, AUFIDIUS:), iambic pentameter constructions.
3	90 (5.1%)	Wikipedia (Notre Dame)	Encyclopaedic prose: factual Wikipedia-style articles about Notre Dame University. Highly formal, third-person, dense noun phrases, institutional and academic vocabulary.

Table 2: Semantic interpretation of the three clusters produced by AHC at $K = 3$.

If a four-cluster solution were desired for theoretical reasons, $K = 4$ would be the next candidate; however, its silhouette score of 0.070843 — a 23% drop from $K = 3$ — provides no quantitative support for the additional split. The silhouette criterion is thus unambiguous: $K = 3$ is optimal.

4.3 Contextualisation Against the Literature

Silhouette scores for text clustering with BERT-family embeddings are structurally low compared to geometric or image datasets, because sentence embeddings reside in a dense, high-dimensional space where cluster boundaries are inherently fuzzy. Petukhova et al. (2024) report silhouette scores for BERT-family models ranging from 0.016 to 0.224 across five diverse text datasets. The score of 0.091923 achieved here falls comfortably within this range — above BERT + k-means on the 20Newsgroups dataset (SS = 0.048) and consistent with reported scores for AHC on comparable text corpora.

This confirms that the result is realistic, well-calibrated, and consistent with state-of-the-art benchmarks. A silhouette score approaching 1.0 would in fact be cause for suspicion: such values typically indicate that a secondary non-linear dimensionality reduction step such as UMAP (McInnes et al., 2018) with `min_dist` = 0 is artificially inflating the metric by compressing points into geometrically tight blobs that are geometrically but not semantically well-separated.

5. Additional Considerations and Limitations

5.1 Cluster Size Imbalance

Cluster 3 (Wikipedia) contains only 90 documents ($\approx$ 5.1% of the dataset). This imbalance is a genuine property of the data rather than an algorithmic artefact. Ward linkage is the linkage criterion most resistant to distortion by cluster size imbalance: its variance-minimisation objective does not favour equal-sized clusters. The small Wikipedia cluster is recovered because its embedding centroid is sufficiently distant from both the Shakespeare and reviews/news centroids in PCA space to justify its existence as a separate group under the silhouette criterion.

5.2 Embedding Model Limitations

MiniLM-L12-v2 produces 384-dimensional embeddings, compared to 768 dimensions for larger models such as `all-mpnet-base-v2`. Larger embeddings capture finer-grained semantic distinctions and may improve the separability of the reviews and news sub-corpora within Cluster 1. However, the assignment constraint of no cloud API dependency and the practical requirement for local inference make MiniLM the appropriate choice. Petukhova et al. (2024) confirm that BERT-family models represent the best open-source option for text clustering, outperforming TF-IDF baselines and performing comparably to much larger open-source LLMs.

References:

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